

Pricing of a novel parametric weather derivative - A Multi-variate Spatial Basket

HKUST

1. Local Risk Exposure - Baseline Stochastic Models Fitted to local Data: $S = f(Y_l(t))$
 - ▶ Weather variables and Climate events - Class of Gauss-Markov processes suitable for indifference pricing measure.
 - ▶ Continuous-time AR(p) model, Seasonal volatility fit.
 - ▶ Generalized Hyperbolic Distribution fit.
 - ▶ Normal Inverse Gaussian fit.
2. Global Traded Index - Futures / Parametric Proxy: $Y_g(t)$
 - ▶ Constructing Real Estate investment portfolio and HDD, CDD, CAT futures index proxy under risk-neutral distribution.
 - ▶ Climate data for Compound Risk Indices.
3. Pricing + Hedging - Spatial Basket: $X(t)$ and $a(t)$ Strategy
 - ▶ Constructing synthetic future index.
 - ▶ Spatio-temporal climate model.

Biovariables and Morphology params

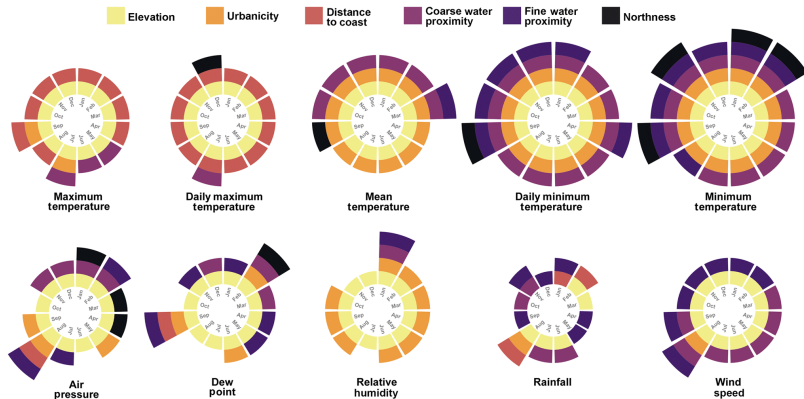


Figure: Reference variables

Real Estate Investment and weather index - Y_I ; Y_g

- Exposure model (since no market portfolio) Real Estate - Estate-wise aggregation: local climate exposure Y_I

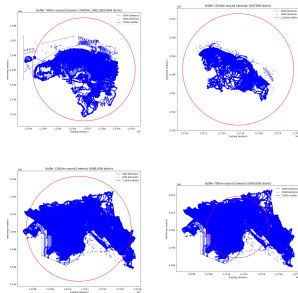


Figure: Intra-District aggregation

Futures on HDD, CDD, CAT, and PRIM - usual traded indices. Multivariate Index strategy based on a linearization based feedback law : get synthetic global climate index Y_g

Pricing and Model Specification

$$F(t, y) \left(\alpha g(t, y) + \frac{1}{2} \frac{\mu^2(y)}{\sigma^2(y)} \right) = \frac{\partial F}{\partial t}(t, y) + GF$$

with terminal condition $F(T, y) = 1$. We decompose the state variable Y by using proxies of y : Y_1 as local risk and Y_2 as global risk (for payoff). Using Feynman-Kac -

$$F(t, Y_1, Y_2) = \mathbb{E}_y \left[\exp \left(- \int_t^T \left(\alpha g(s, Y_2) + \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} \right) ds \right) \right].$$

E_y - over joint distribution Y_1 and Y_2 s.t. inf generator G accounts for coupling terms. with $V(t, x, y) = -e^{-\alpha x} F(t, Y_1, Y_2)$

General Solution with non-coupled dynamics (Carmona 2005) -

$$p = \frac{1}{\alpha} \ln \left(\frac{\mathbb{E}_y \left[\exp \left(- \int_0^T \frac{1}{2} \frac{\mu^2(y_s)}{\sigma^2(y_s)} ds \right) \right]}{\mathbb{E}_y \left[\exp \left(- \int_0^T \left(\alpha g(s, \hat{Y}_s) + \frac{1}{2} \frac{\mu^2(y_s)}{\sigma^2(y_s)} \right) ds \right) \right]} \right)$$

Pricing and Model Specification

Under independent non-traded risk dynamics Y_1 and Y_2 or when coupling effect is negligible or can be treated as a perturbation to the independent case -

With terminal condition $F(T, Y_1, Y_2) = 1$ -

$$F(t, Y_1, Y_2) \left(\alpha g(t, Y_2) + \frac{1}{2} \frac{\mu^2(Y_1)}{\sigma^2(Y_1)} \right) = \frac{\partial F}{\partial t}(t, Y_1, Y_2) + \mathcal{L}_{Y_1} F + \mathcal{L}_{Y_2} F$$

$$\begin{aligned} F(t, Y_1, Y_2) &= \mathbb{E}_{Y_1, Y_2} \left[\exp \left(- \int_t^T \left(\alpha g(s, Y_2(s)) + \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} \right) ds \right) \right] \\ &= \mathbb{E}_{Y_1} \left[\exp \left(- \int_t^T \frac{1}{2} \frac{\mu^2(Y_1(s))}{\sigma^2(Y_1(s))} ds \right) \right] \times \mathbb{E}_{Y_2} \left[\exp \left(- \int_t^T \alpha g(s, Y_2(s)) ds \right) \right] \end{aligned}$$

Trading strategy: $\phi_t = \frac{\mu(Y_1)}{(\sigma^2(Y_1)\alpha)}$

Pricing and Model Specification

Coupled case with non-0 correlation b/w local risk and global risk -

$$p = \frac{\ln \mathbb{E} \left[e^{(1-\rho^2)\Psi(Y_2(T))} \frac{dQ}{dP} \right]}{\alpha(1-\rho^2)}, \quad \frac{dQ}{dP} = e^{-\frac{1}{2} \int_0^T (\mu_1^2/\sigma_1^2) d\tau + \int_0^T \frac{\mu_1}{\sigma_1} dW_t^S}$$

$$\phi_t = \frac{\rho}{\sigma(Y_1)} a(Y_2) p' + \frac{\mu(Y_1)}{\sigma(Y_1)^2 \alpha}.$$

where -

$$dS_t = \mu(Y_1, t) dt + \sigma(Y_1, t) dW_t^S$$

$$dY_t = b(Y_2, t) dt + a(Y_2, t) (\rho dW_t^S + \sqrt{1-\rho^2} dW_t^{S\perp})$$

$$W^{S\perp} = \frac{1}{\sqrt{1-\rho^2}} W^Y - \frac{\rho}{\sqrt{1-\rho^2}} W^S$$

Temporal Dynamics of State Variables

Temperature example - for basis risk as the source of market incompleteness

$$dT(t, x) = d\Lambda(t, x) - \alpha(x)(T(t, x) - \Lambda(t, x)) dt + \sigma(t, x) dW(t, x)$$

- ▶ $W(t, \cdot)$: $L^2(D)$ -valued Wiener process
- ▶ Continuous spatial covariance function $q(x, y)$, strictly positive definite and symmetric.
- ▶ Given weather index $I(x_i)$ at locations $x_1, \dots, x_n$.
- ▶ Optimal strategy $a(t)$ minimizing variance of error in replicating $I(y)$ at location y using spatial correlation and geometry of locations.
- ▶ Risk Neutral Time-Dynamics of Index Futures Prices;

$$dF_{\text{index}}(t, \tau_1, \tau_2) = \sigma(t) \int_{\tau_1}^{\tau_2} e_1' \exp(A(s - t)) e_p \Phi$$

Φ is the cumulative standard normal distribution

Fitting the Model

- Explicit solution of $X(t)$:

$$X(s) = \exp(A(s - t))x + \int_t^s \exp(A(s - u))e_p\sigma(u) dB(u)$$

- Temperature $T(t)$ is defined as:

$$T(t) = \Lambda(t) + X_1(t)$$

- $X_1(t)$: CAR(p) model, $\Lambda(t)$: seasonality function

1. Fit seasonal function $\Lambda(t)$ with least squares.
2. Fit AR(p) model on deseasonalized temperatures.
3. Fit seasonal volatility $\sigma(t)$ to residuals.

Contract Payoff and Pricing

Utility with a derivative written on the underlying process Y - akin to Asian structure and $K = 0$

$$f_1(Y) = \left(\int_0^T Y_t dt - K \right)^+; f_2(Y) = \left(\int_0^T \mathbb{1}_{\{Y_t > \epsilon\}} dt - K \right)^+,$$

$$f(Y) = \int_0^T h(Y_s) ds, \quad 0 \leq t' \leq t \leq T.$$

- ▶ Numerical solution of underlying optimization problem analysed through its dual under HJB (David)
- ▶ Explicit closed form solution to HJB: RN measure vs historical measure vs Indifference measure (Musiela)

Spatial Hedging - Weight Matrix - Global index Y_g

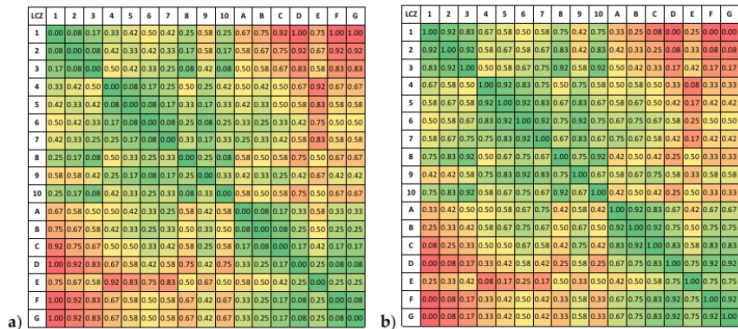


Figure 3. (a) Dissimilarity and (b) similarity metric for LCZ classes (Appendix A in [4]).

Figure: LCZ (Local Climate Zone) based wights matrix reference

Spatial Hedging - Weight Matrix - Global index Y_g

- ▶ Model with no spatial dependency in Λ , α , and σ .
- ▶ Spatial dependency modeled by a spherical correlation function (Benth 2012)

$$q(x, y) = 1 - \frac{3}{2} \frac{|x - y|}{\gamma} + \frac{1}{2} \frac{|x - y|^3}{\gamma^3}$$

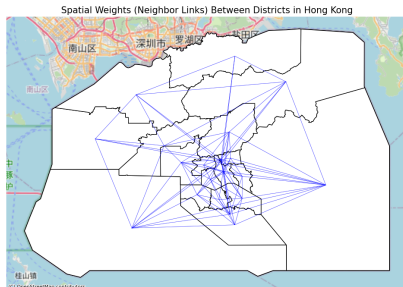


Figure: Rent Roll based